

ASSIGNMENT 17

ROLL NO:CSB24083

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Registration_service.py

```
import re

class InvalidEmailError(ValueError):
    """Raised when the email is invalid."""
    pass

class UnderageError(RuntimeError):
    """Raised when the user's age is below 18."""
    pass

class RegistrationService:

    EMAIL_PATTERN = r'^[A-Za-z0-9._%+-]+@[A-Za-z0-9.-]+\.[A-Za-z]{2,}$'

    def register_user(self, email: str, age: int) -> bool:

        # Internal invariant check
        assert email is not None, "System invariant violated: email cannot be None"

        # Validate email
```

```
    if not email or email.strip() == "":
        raise InvalidEmailError("Email cannot be empty.")

    if not re.match(self.EMAIL_PATTERN, email):
        raise InvalidEmailError("Invalid email format.")

    # Validate age
    if age < 18:
        raise UnderageError("User must be at least 18 years
old.")

    return True
```

test_registration_service.py

```
import pytest
from registration_service import (
    RegistrationService,
    InvalidEmailError,
    UnderageError
)

@pytest.fixture
def service():
    return RegistrationService()

def test_successful_registration(service):
    assert service.register_user("user@example.com", 22) is True

def test_empty_email(service):
```

```
with pytest.raises(InvalidEmailError):
    service.register_user("", 25)

def test_invalid_email_format(service):
    with pytest.raises(InvalidEmailError):
        service.register_user("invalid-email", 25)

def test_underage_user(service):
    with pytest.raises(UnderageError):
        service.register_user("user@example.com", 16)

def test_none_email_assertion(service):
    with pytest.raises(AssertionError):
        service.register_user(None, 20)
```

OUTPUT

```
rootdir: F:\AP projects
collected 5 items

test_registration_service.py::test_successful_registration PASSED [ 20%]
test_registration_service.py::test_empty_email PASSED [ 40%]
test_registration_service.py::test_invalid_email_format PASSED [ 60%]
test_registration_service.py::test_underage_user PASSED [ 80%]
test_registration_service.py::test_none_email_assertion PASSED [100%]

===== 5 passed in 0.80s =====
```